

Unemployment Challenges in Africa Dashboard Analysis

SUMMARY

The rate of unemployment in Africa across 61 countries in the year 2015 that this dataset was able to provide without discrepancies are; **Dijibouti** topping that chart with **47.97%** unemployment rate by average, **11.22% Avg female** unemployment, **8.14% Avg male** unemployment and **15.29% Avg unemployment**. The quality of education and access to electricity seem to reduces the rate of unemployment and vice versa.

Key Findings

1. Gender Gap

Gender gap is the difference between female unemployment and male unemployment. And this is calculated to check the availability and potential job gap for woman.

Female are shut out of the job market compared to male in various Africa countries with **Sudan** topping the chat with 16.7% gender gap,

Outliers were also discovered in countries like Niger, Ghana, Guinea, Liberia, Chad, Burundi, Togo, Congo, Sierra Leone have negative numbers which can be identified as men having low employment rate there.

2. Electricity Access

Following the analysis made in countries, countries with broader electricity access generally show lower unemployment levels, suggesting infrastructure development may support economic participation. Electricity access aligns with stronger labor outcomes.

3. Education Access

Following the analysis made in 2015, I would suggest that more investment be directed to the educational system to reduce the rate of unemployment. Education investment shows moderate negative relationship with unemployment.

4. Unemployment vs Total limited company

Following the analysis made in 2015, after merging with the unemployment gender table only 14 countries were matched excluding duplicates, analysis shows that the number of companies in a country does not affect the unemployment rate as we have countries with high number of employment rate – high number of local limited companies, indicating that the rate of unemployment is not connected to the number of local limited companies, but could be other reason such as poor payment, poor skills, inexperienced adults, few number of young people.

5. Youth Strategies.

Countries with operational youth strategies perform relatively better than the others (adopted, developing and no strategy).

Challenges Encountered

- Dataset 1 is a single-year snapshot no trend analysis possible. The unemployment by gender dataset covers only 2015 for all 61 African countries. This makes it impossible to observe whether unemployment improved or worsened over time from this dataset alone. Any conclusions about unemployment levels reflect one point in time and cannot show direction of change. **A dataset spanning 2010–2022 would have been needed for trend analysis.**
- Future collection should prioritize standardized time-stamps to avoid high-volume manual cleaning.
- Data discrepancies, missing records, I wasn't able to merge a master sheet because the dataset has different years and different countries which made it difficult to create one without null and blank spaces.
- Correlation does not imply causation. Any relationships found between variables for example between education spend and unemployment, or electricity access and unemployment, are correlational observations, not causal proof. Recommendations based on these correlations should be treated as hypotheses to be tested with further research.

Recommendation

- Following the analysis made in 2015, **I would suggest that more investment be directed to the educational system to reduce the rate of unemployment.** Education investment shows moderate negative relationship with unemployment.
- Following the analysis made in countries, Countries with broader electricity access generally show lower unemployment levels, **suggesting infrastructure development may support economic participation.** Electricity access aligns with stronger labor outcomes
- Female unemployment remains consistently higher among countries in 2015. **Women should be given more opportunity in the labor market.**